

XEROJ ULGASAN

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EDUCATION

QUEZON CITY UNIVERSITY

Bachelor of Science in Information Technology

San Bartolome, Q.C

2023 - Expected 2027

ASIAN INSTITUTE OF COMPUTER STUDIES

Science, Technology, Engineering, and Mathematics

Commonwealth, Q.C

2021 - 2023

UNIVERSITY PROJECTS

ERBriwan

January 2026 - April 2026

- Engineered an IoT-based emergency response platform integrating 3 core components (ESP32 device, cloud API, and mobile application) to enable real-time distress reporting and location tracking together with FreeRTOS to maximize performance.
- Architected a scalable Node.js REST API utilizing Express and PostgreSQL to route emergency alerts; optimized query performance to maintain sub-200ms response times for 100+ concurrent IoT device connections.
- Engineered a mobile application featuring real-time incident monitoring, GPS-based visualization, and push notification delivery to improve emergency coordination with near real-time (~1s) updates.
- Implemented an end-to-end emergency alert workflow that consolidated multiple communication channels into a single-button trigger, significantly reducing reporting time from 10-15 seconds into nearly 1 second during critical incidents.

BarangayEase

January 2026 - April 2026

- Built a full-stack community service management platform supporting complaint submission, document requests, and case tracking within a centralized system, reducing barangay paperwork by 60%.
- Developed a secure authentication and authorization system using JWT, RBAC, and Row-Level Security for 2 user groups, protecting sensitive PII and ensuring data integrity across administrative workflows.
- Unified resident-facing and administrative portals into a single platform, eliminating duplicate record handling and improving request visibility by 30%+.
- Deployed containerized background services using Docker on AWS EC2 to automate core processes and reduced manual overhead by 40% through scheduled CRON tasks and automated notification triggers.

QCU-EcoCharge

August 2025 - December 2025

- Built an IoT-powered solar monitoring system integrating multiple environmental and energy sensors for real-time performance analysis, supporting 50 concurrent users with sub-5-second latency.
- Established backend services using Node.js to collect, store, and manage electricity generation data, device status, and historical energy records, enabling users to track performance trends over time.
- Created a React dashboard capable of visualizing real-time and historical energy metrics to support renewable energy monitoring with less than 2 second of loading time at 50 concurrent users.
- Engineered a decoupled 4-layer system architecture (Hardware, API, Data, UI) using MVC principles, ensuring modularity and facilitating independent testing of hardware integration and frontend visualization.

PondPal

January 2025 - June 2025

- Delivered a smart aquaculture monitoring platform capable of tracking 4 key water-quality parameters including temperature, pH, turbidity, and TDS levels.
- Implemented a real-time pond monitoring dashboard with automated alerts for abnormal environmental conditions, notifying users within 1 minute of anomaly detection.
- Designed a historical data management system supporting trend analysis and long-term water quality assessment.
- Enabled centralized monitoring of multiple ponds and sensor nodes through a unified interface, improving farm management efficiency by ~30%.

QCITS

August 2024 - December 2024

- Built a seat reservation and monitoring system using VB.NET to manage auditorium bookings capable of handling 450 event participants.

ADDITIONAL

Programming Languages

JavaScript, Java, Python, C#, C/C++, PHP, SQL, VB.NET

Frameworks & Libraries

React, Node.js, Express, Spring Boot

Databases

MySQL, Firebase, Oracle, PostgreSQL

Embedded & IoT

ESP32, Arduino, Sensor Integration, Microcontrollers

Tools

Git, GitHub, Postman, Docker

Concepts

REST APIs, System Design